

Undercut Analyser: Predicting Undercut Opportunities From Historic Lap Time Data in Formula One



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PROBLEM & AIM

An undercut pits the attacking driver before a competitor ahead, using fresher-tyre performance to try to gain track position once the target pits themselves [1]. **The decision is a relative-performance problem: the time gained must be large enough to overcome the initial gap.** Aim: develop, implement and validate a deterministic model using historic F1 data and configurable engineering parameters.



MODEL FORMULATION

$$L_d(i) = B_d + C_{cd(i)} + D_{cd(i)}(a_d(i)) + W_c(i) + R_d(i) + Q_d(i)$$

$$G(m) = G_0 + [T_{att}(m) - T_{tgt}(m)]$$

$L_d(i)$ - Predicted lap time for driver d on lap i
 B_d - Baseline lap time for driver d , calculated from the reference lap time and driver/car pace offset.
 $C_{cd(i)}$ - Lap time offset for compound c fitted to driver d on lap i .
 $D_{cd(i)}$ - Lap time loss due to degradation of the fitted tyre compound.
 $a_d(i)$ - Tyre age for driver d on lap i .
 $W_c(i)$ - Configurable lap time penalty for compound c from tyre warm-up applied following a tyre change.
 $R_d(i)$ - Configurable lap time penalty to model traffic interaction.
 $Q_d(i)$ - Configurable lap time penalty to represent pit stop loss.
 $T_d(m)$ - Cumulative predicted time of attacking (att) and target (tgt) driver after m predicted laps.
 G_0 – Initial attacker-to-target gap at the decision point.

+ve G = attacker behind. ΔG isolates the time generated by the undercut from whether it changes track position.

HISTORIC DATA & PARAMETERISATION

Historic race data is acquired from public sources online, and decision points created based on each driver’s cumulative time at sector line two. The model only uses parameters available at the decision point and does not use any information from later in the race. The model operating window is designed around standard green-flag racing. It is not designed to operate in changing weather conditions or handle unexpected racing conditions such as safety cars.

The prediction architecture is causal and does not use future race state, but historic validation parameters were retrospectively derived from full-race data to provide representative engineering inputs.

References: [1] Formula 1, “Undercut vs Overcut: Why Tyre Strategy Was So Finely Poised in Monaco” 2021. [2] OpenF1 Contributors, “OpenF1,” 2024, doi: 10.5281/zenodo.11241279. [3] S. Razavi and H. V. Gupta, “What Do We Mean by Sensitivity Analysis?” Water Resour. Res., vol. 51, no. 5, pp. 3070–3092, 2015.

APPLICATION

A **C#/.NET WPF application** was written to run predictions on single drivers and scan across full race distances and multiple drivers. Data is retrieved from the **OpenF1 API** [2]. Results are displayed on a race-trace visualisation which exposes intermediate model behaviour. Prediction logic was kept separate from data retrieval and presentation to assist verification and allow easy future expansion.

SOFTWARE VERIFICATION

Automated testing was written to verify the software implementation, with **460 tests** and **37 research-focused cases** giving **95.6% core line coverage** and **89.7% branch coverage**. Independent calculations matched software outputs, and tests cover lap-time terms, pit sequence, tyre-age progression, sign convention, $G(m)$ and $\Delta G(m)$, classification boundaries, limiting cases and deterministic repeatability.

MODEL VALIDATION

Historic cases were selected to include successful and unsuccessful undercut attempts, one- or two-lap responses, and different tyre-state transitions. The model matched **3 of 4 classifications** but **4 of 4 directions of change**. Belgium provided the strongest immediate numerical agreement with **0.024 s error at $n+o$** ; Hungary predicted the correct direction of relative gain but underestimated its magnitude, resulting in an incorrect classification. The discrepancy was partly attributed to real-world factors such as differences in stationary pit stop time.

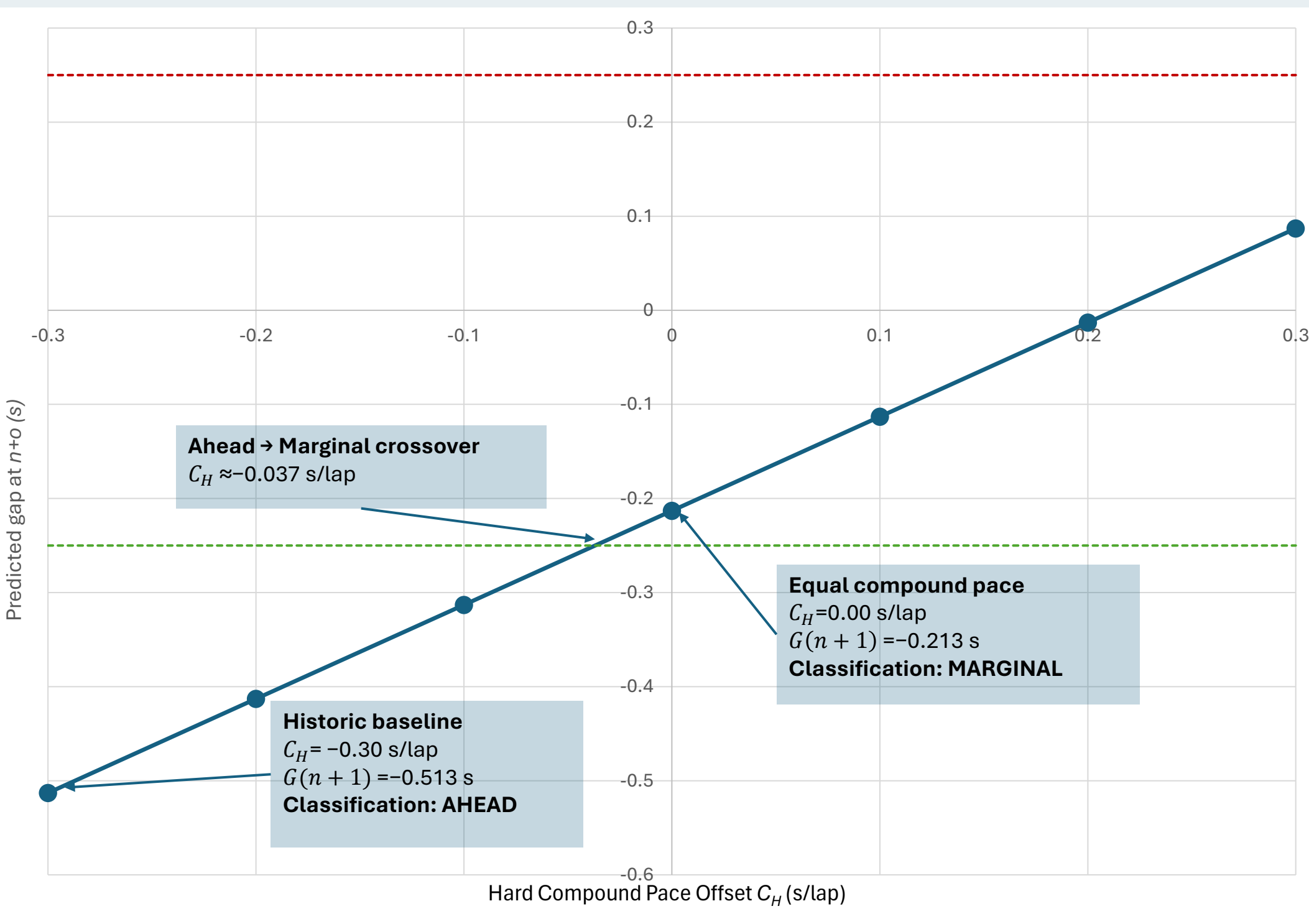
SENSITIVITY ANALYSIS

One-at-a-time sweeps [3] tested the impact of compound pace, degradation, target response offset, and relative driver/car pace. **Italy 2024 showed particularly strong sensitivity to compound pace offset** (shown in graph to the right): varying Hard-compound pace alone can move the result into Marginal; the **Ahead-Marginal crossover occurs at $C_H \approx -0.037\text{s/lap}$** . Analysis suggested that parameter influence is scenario-dependent; therefore, the research does not claim to create a universal sensitivity ranking of parameters and their influence.

Historic Validation Cases

Race	Predicted	Observed	Error at $n + o$
Hungary 2023 - Norris → Piastri (L17)	Behind	Ahead	1.148 s
Belgium 2024 - Leclerc → Hamilton (L25)	Behind	Behind	0.024 s
Italy 2024 - Norris → Leclerc (L14)	Ahead	Ahead	1.615 s
Japan 2025 - Russell → Leclerc (L19)	Behind	Behind	0.189 s

Sensitivity Analysis – Compound Offset (Italy 2024)



CONCLUSIONS

- The model consistently identified whether pitting first generated a **relative gain or loss across the four historic cases**.
- **Exact realised gaps were less reliable** because fixed deterministic inputs cannot reproduce every short-term race effect.
 - **Initial gap and undercut gain must be considered separately:** a beneficial undercut can still leave the attacker behind.
 - **Important omitted or simplified effects** include dynamic traffic, unequal pit-stop execution, nonlinear tyre behaviour and stochastic lap-to-lap variation.
 - **The deterministic structure remains inspectable:** the contribution of each model term can be traced rather than hidden inside an unexplained recommendation.

A transparent deterministic model can provide useful engineering analysis of undercut opportunities in Formula One races within a defined operating domain. It is best treated as **an analysis tool to support an engineer’s decision-making process**, not a complete race strategy simulator, or an autonomous pit stop recommendation.

